Equations

An equation is something of the form

one expression = another expression

"Statement of Equality"

These expressions are typically algebraic

e.g. $x^2 - 5 = 4x$ (we'll come back to this example)

quadratic equation

The solutions/roots of an equation are numbers you can plug in for the variables that make the equation true

We break down equations into 3 classes based on their solution sets (set of all solutions):

quation is of the identity type if it is
ue for every value in its domain

e.g. $x+3=x+3$, $\frac{5}{x-1}=\frac{5}{x-1}$, $x^2+y=x^2+y$

true for all $x \neq 1$

"$0=0$" type equations

An equation is of the conditional type if
~~e.g.~~
it is true only for some values in its domain

e.g. $x+3=1 \xrightarrow{-1} x=2$, $\frac{5}{x-1}=1 \rightarrow x=6$, $x^2+y=1 \rightarrow y=1-x^2$

"$x=0$" type equations

An equation is of the contradiction type if it is not true for any value in its domain

e.g. $x+3=x+3+1$, $\cancel{\frac{5}{x-1}=\frac{5}{x-1}+4}$, $\frac{5}{x-1}=0$,

$x^2+\cancel{y}=\cancel{y}-1 \rightarrow x^2=-1$

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equations are <u>equivalent</u> if they have exactly the same solution sets

e.g. $\underbrace{x = 0 \xrightarrow{\times 2} 2x = 0 , \; x^2 = 0}_{\text{equivalent}}$

$\underbrace{x = 1 , \; 2x = 1 , \; x^2 = 1}_{\text{all inequivalent}}$

<u>Solve</u>: $\boxed{6x - 7 = 2x + 5}$

$\Rightarrow 6x = 2x + 5 + 7$

$\Rightarrow 6x = 2x + 12$

$\Rightarrow 6x - 2x = 12$

$\Rightarrow 4x = 12$

$\Rightarrow \boxed{x = 3}$

equivalent

$$(3x-2)^2 = (x-5)(9x+4)$$

$$\cancel{9x^2} - 12x + 4 = \cancel{9x^2} + 4x - 45x \mp 20$$

$$-12x + 4 = -41x \mp 20$$

$$29x = \cancel{16} - 24$$

$$x = \frac{-24}{29}$$

Solve:

$$\frac{3x}{x-2} = 1 + \frac{6}{x-2}$$

$$x - 2 \neq 0$$

$$x \neq 2$$

Domain

$\times(x-2)$ on both sides

$$3x = \left(1 + \frac{6}{x-2}\right)(x-2)$$

$$3x = 1(x-2) + \frac{6}{\cancel{(x-2)}}\cancel{(x-2)}$$

$$3x = x - 2 + 6$$

$$\Rightarrow 3x = x + 4 \Rightarrow 2x = 4 \Rightarrow \boxed{\cancel{x = 2}}$$

No solutions

$$x^2 - 16 = 0$$

$$\Rightarrow x^2 = 16, \quad x = \pm\sqrt{16}$$

$$x = +4 \text{ or } -4$$

Factor? $x^2 - 16 = (x-4)(x+4)$

$$\underbrace{(x-4)}\underbrace{(x+4)} = 0$$

one of these has to be 0

Solve: $\dfrac{1}{x+4} + \dfrac{3}{x-4} = \dfrac{3x+8}{x^2-16}$

domain $x \neq 4, -4$

$$\frac{1}{x+4} + \frac{3}{x-4} = \frac{3x+8}{(x+4)(x-4)}$$

$$\frac{x-4}{\cancel{(x+4)(x-4)}} + \frac{3(x+4)}{\cancel{(x+4)(x-4)}} = \frac{3x+8}{\cancel{(x+4)(x-4)}}$$

$$x - 4 + 3(x+4) = 3x + 8$$

$$x + 8 = 8 \Rightarrow x = 0$$

w that the equation is identity:

$$(4x-3)^2-16x^2=9-24x$$

$$(a\pm b)^2=a^2\pm 2ab+b^2$$

$$(4x)^2-2(4x)(3)+3^2$$

$$16x^2-24x+9$$

$$\cancel{16x^2}-\cancel{24x}+\cancel{9}-\cancel{16x^2}=9-\cancel{24x}$$

$$\boxed{0=0}$$

that the equation

is identity:

Domain $x \neq 0$

$$\frac{3x^2+8}{x} = \frac{8}{x} + 3x$$

$$\frac{3x^2+8}{x}$$

$$\frac{3x^2}{x} + \frac{8}{x}$$

$$= \frac{3x}{\cancel{x}} + \frac{8}{x}$$

$$\frac{8}{x} + \frac{3x}{1} \quad LCD = x$$

$$= \frac{8}{x} + \frac{3x \cdot x}{x}$$

$$= \frac{8+3x^2}{x}$$

the following equations equivalent?

$$\frac{8x}{x-7} = \frac{72}{x-7}$$

domain

$x \neq 7$

$x - 3 = 6$

$x = 9$

$$\frac{8x}{x-7} = \frac{72}{x-7} \cdot (x-7)$$

because $x \neq 7$

$8x = 72$

$x = 9$

e

...the following
...quations equivalent?

$$\frac{8x}{x-7} = \frac{56}{x-7}$$

domain

$x \neq 7$

$$\frac{8x}{x-7} = \frac{56}{x-7} \quad \cdot(x-7) \text{ valid since } x \neq 7$$

$8x = 56$

$x = 7$

contradiction type equation

$2x - 4 = 10$

$2x = 14$

$x = 7$